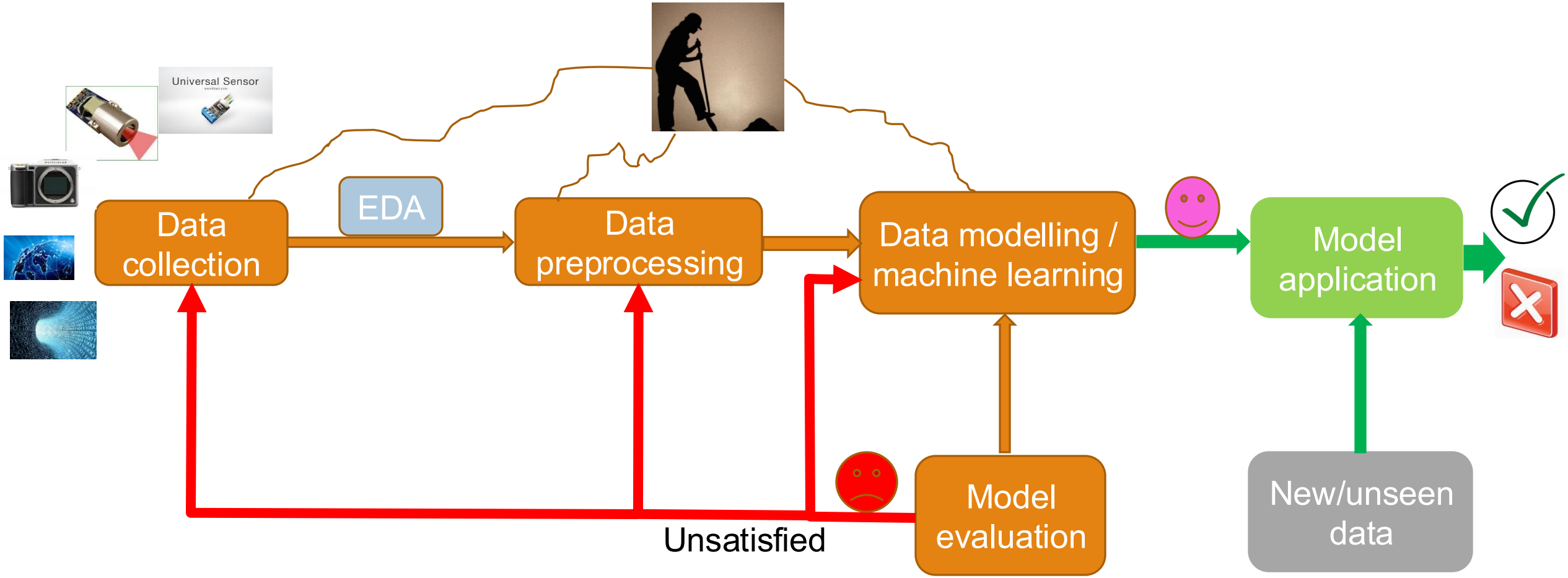
2. Data Pre-processing

Applied Machine Learning

Yuhua Li
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Logical Process (Pipeline) of Machine Learning



Contents

- Introduction
- Data inspection
- Transformation
- Dealing with missing data
- Encoding categorical features
- Imbalanced data
- Data reduction

Introduction

- Why is data pre-processing Important?
  - No quality data, no quality modelling results!
  - Quality decisions must be based on quality data

    e.g., duplicate or missing data may cause incorrect

    or even misleading statistics.
  - *Data preparation, cleaning, and transformation comprises the majority of the work in a machine learning application.*

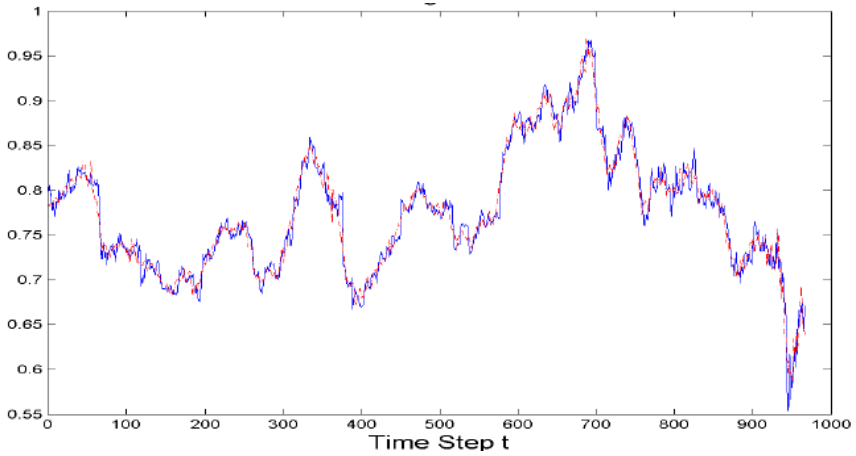
Introduction

- Data Validation
  - Data in the real world is dirty
  - Incompleteness
    - missing attribute values, lack of certain attributes of interest, or containing only aggregate data. E.g. Postcode = “”
  - Noise
    - Data may contain erroneous or anomalous values. These are referred to as outliers. E.g. age = -2
  - Inconsistency
    - Sometimes data contains discrepancies.

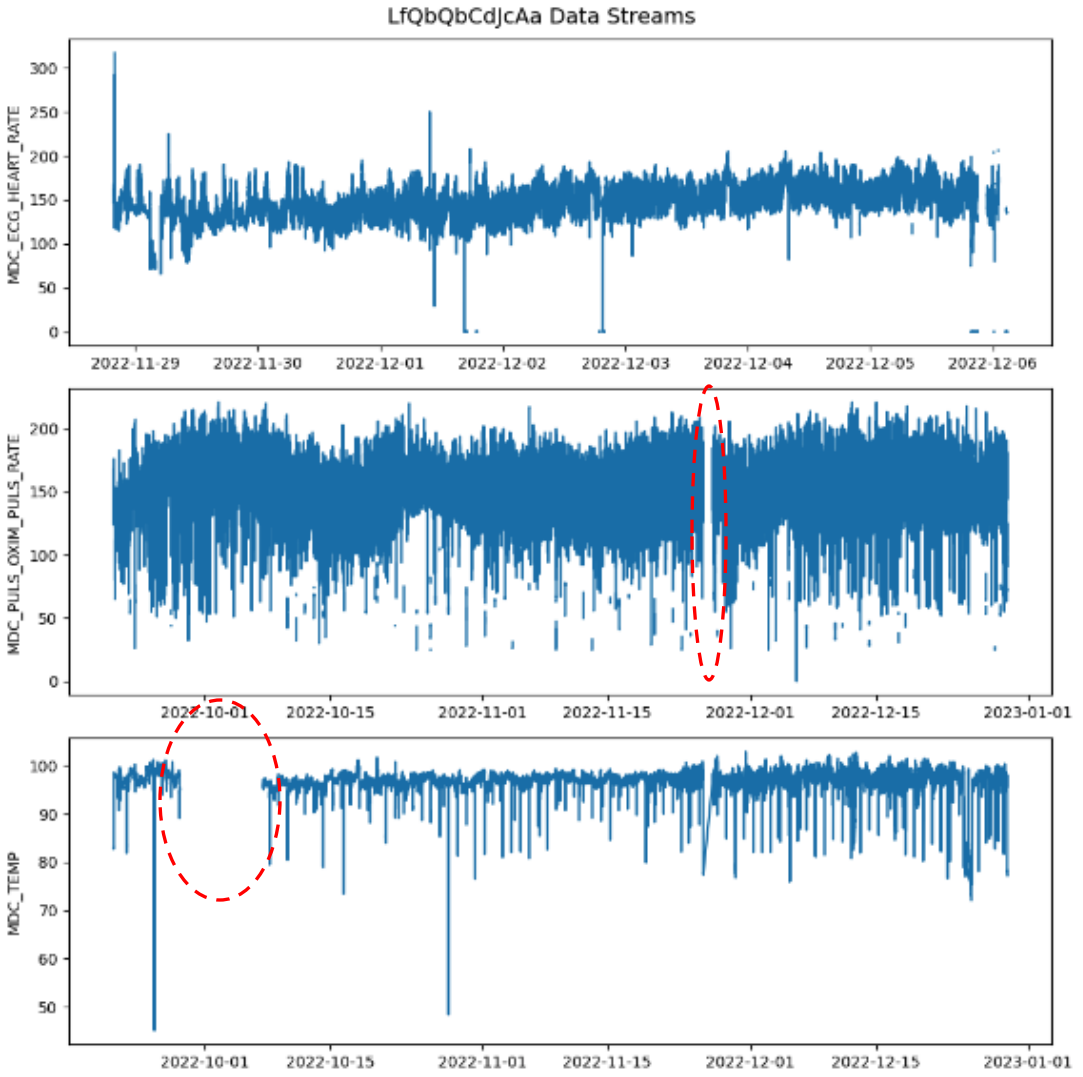
      e.g., Age=“52” Birthday=“01/01/1991”

Noisy Data

- Noise: random error or variance in a measured variable.



- Incorrect attribute values may due to
  - faulty data collection instruments
  - data entry problems
  - data transmission problems
  - etc
- Other data problems which requires data cleaning
  - duplicate records, incomplete data, inconsistent data

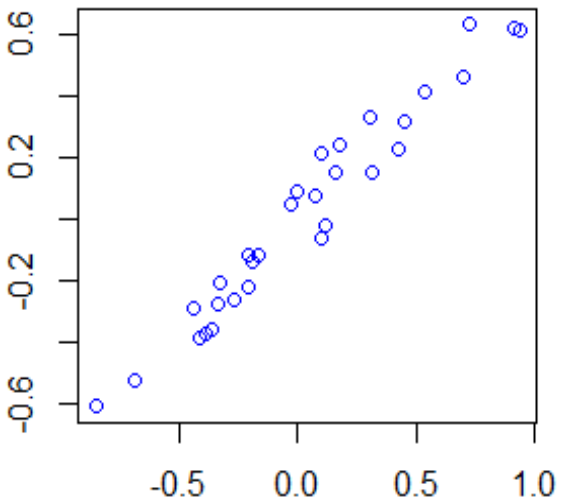


Contents

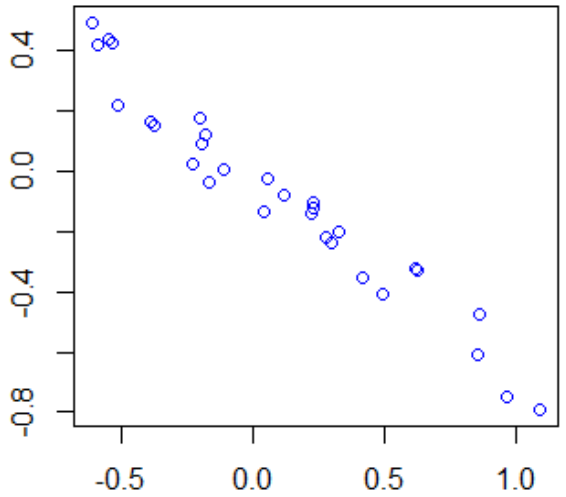
- Introduction
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Data Inspection - Exploratory data analysis

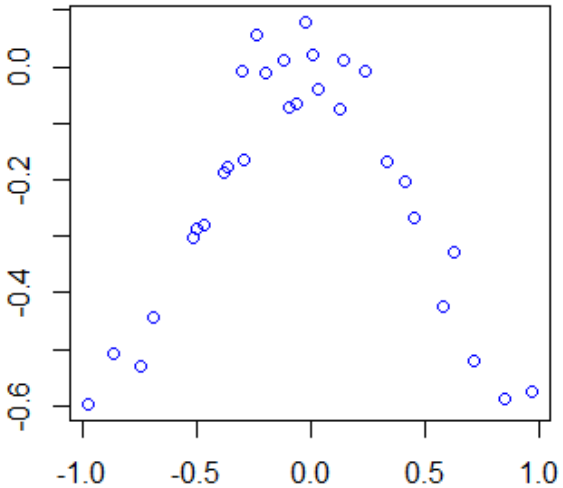
- Scatter plot
  - Provides a first look at bivariate data to see clusters of points, outliers, etc
  - Each pair of values is treated as a pair of coordinates and plotted as points in the plane



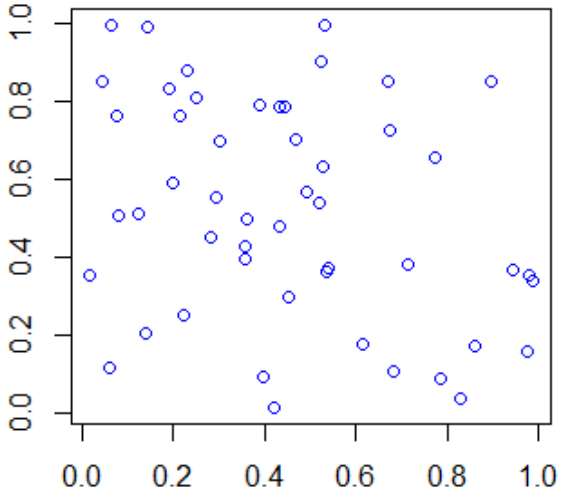
Positively correlated



Negative correlated



Nonlinear correlated



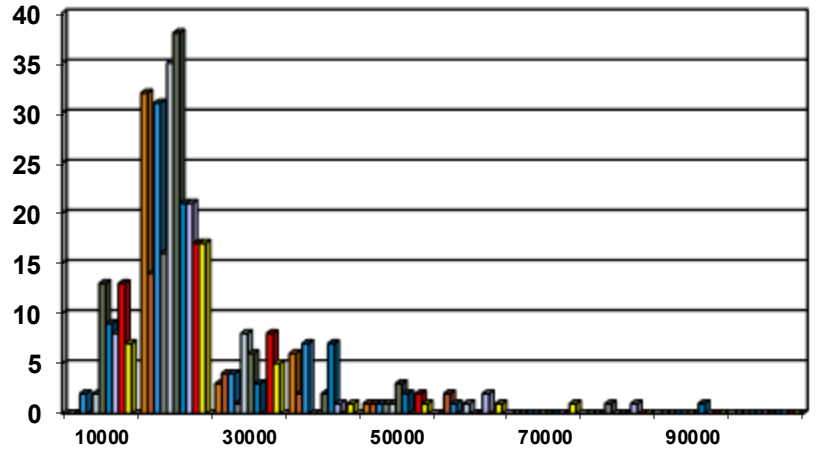
Uncorrelated

Correlation coefficient ϵ [-1, 1],

$$r_{xy} = \frac{\sum(x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum(x_i - \bar{x})^2 \sum(y_i - \bar{y})^2}}$$

Data Inspection

- Histogram
  - Graph display of tabulated frequencies, shown as bars. x-axis are values, y-axis represent frequencies
  - It shows what proportion of cases fall into each of several categories
  - The categories are usually specified as non-overlapping intervals of some variable. The categories (bars) must be adjacent

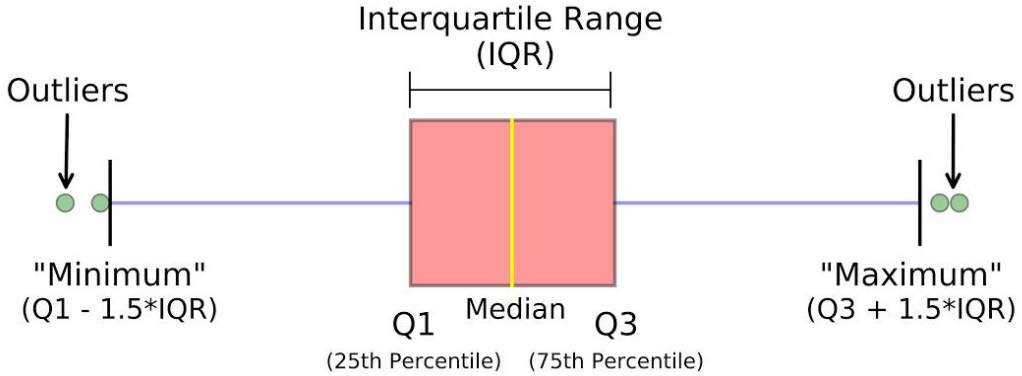


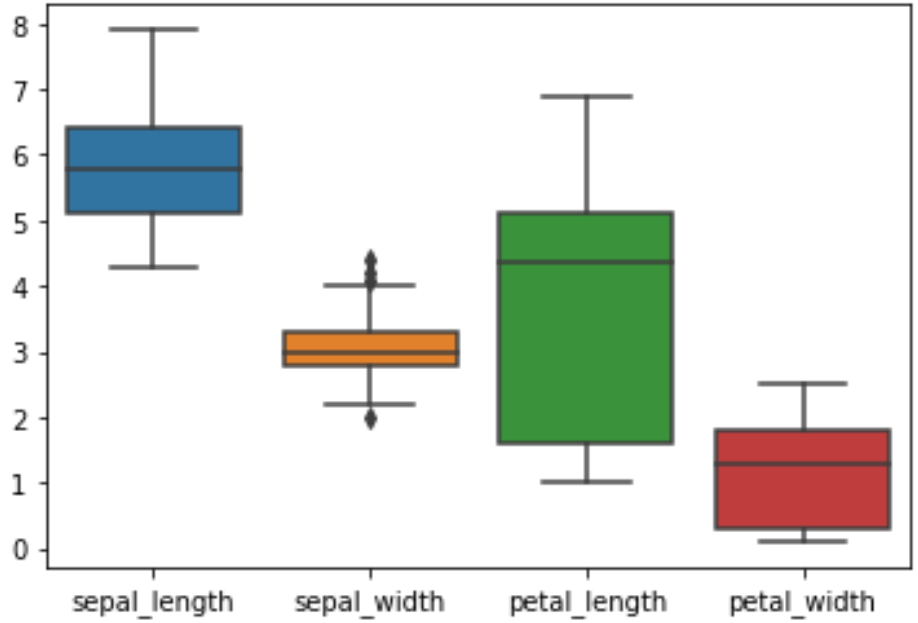
Fname	Lname	Salary
John	Smith	30000
Franklin	Wong	40000
Joyce	English	80000
Ramesh	Narayan	38000
James	Borg	55000
Jennifer	Wallace	43000
Ahmad	Jabbar	25000
Alicia	Zeala	25000

More data ⇣ ⇣ ⇣

Data Inspection

- **Five-number summary** of a distribution
  - Minimum, Q1, Median, Q3, Maximum
- **Boxplot**
  - Data is represented with a box
  - The ends of the box are at the first and third quartiles, i.e., the height of the box is IQR
  - The median is marked by a line within the box
  - Whiskers: two lines outside the box extended to (Q1 – 1.5*IQR) and (Q3 + 1.5*IQR)
  - Outliers: points beyond whiskers, plotted individually



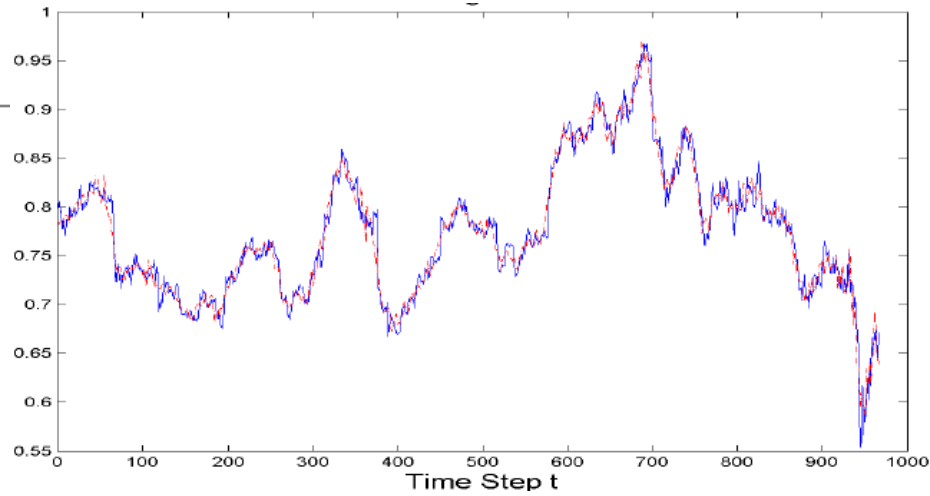


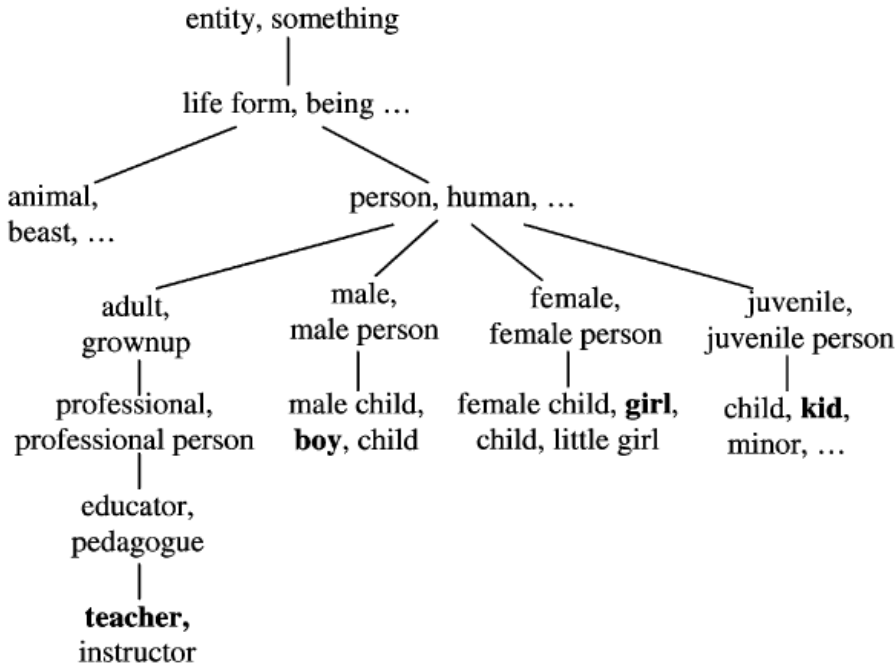
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Data Transformation

- Smoothing: remove noise from data (binning, clustering, regression)
- Scaling: scaled to fall within a small, specified range such as –1.0 to 1.0, 0.0 to 1.0 or $\mathcal{N}(0, 1)$
- Attribute/feature construction
  - New attributes constructed / added from the given ones
- Aggregation: summarization or aggregation operations apply to data
- Generalization: concept hierarchy climbing
  - Low level/ primitive/raw data are replace by higher level concepts





Feature Scaling (Data Normalization)

- The range of values of raw data varies widely, scaling makes all features have a same range
  - Necessary for distance/similarity based ML methods
  - Faster convergence for gradient descent learning algorithms
  - Appropriately penalizing coefficients in loss functions involving regularisation
- min-max normalization
  - For a variable *X*, $x \in X$, the range of X is normalized from its original range $[min_X, max_X]$ to new range $[a, b]$

$$x' = a + \frac{b - a}{max_X - min_X}(x - min_X)$$

E.g., Normalising dataset (9, 22, 14, 2, 11) to new range [-1, 1] becomes (-0.3 1.0 0.2 -1.0 -0.1)

sklearn.preprocessing.MinMaxScaler sklearn.preprocessing.minmax_scale

- z-score normalization (Standardization)

$$x' = \frac{x - mean_X}{std_X}$$

sklearn.preprocessing.StandardScaler

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Dealing with missing data

Titanic data

Survived	Pclass	Sex	Age	SibSp	Parch	Fare	Cabin	Embarked
0	3	male	22	1	0	7.25	NaN	S
1	1	female	38	1	0	71.2833	C85	C
1	3	female	26	0	0	7.925	NaN	S
1	1	female	35	1	0	53.1	C123	S
0	3	Male	35	0	0	8.05	NaN	S
0	3	Male	NaN	0	0	8.4583	NaN	Q
0	1	Male	54	0	0	51.8625	E46	S
0	3	Male	2	3	1	21.075	NaN	S
1	3	Female	27	0	2	11.1333	NaN	S
1	2	Female	14	1	0	30.0708	NaN	C

Dealing with Missing Data

- Elimination
  - Discard all records for which the values of one or more attributes are missing. In the case of supervised learning, it is essential to eliminate a record if the value of the target attribute is missing.
- Inspection
  - Alternatively, one may opt for an inspection of each missing value, carried out by experts in the application domain, in order to obtain recommendations on possible substitute values.

Dealing with Missing Data

- Identification
  - As a third possibility, a conventional value might be used to encode and identify missing values, making it unnecessary to remove entire records from the given dataset.
- Missing-data imputation
  - It is the process of replacing missing data with substituted values
  - Several criteria exist for the automatic replacement of missing data, although most of them appear somehow arbitrary. For instance, missing values of an attribute may be replaced with the mean of the attribute calculated from the remaining observations.

Dealing with Missing Data

- Get rid of the corresponding instances.
- Get rid of the whole attribute.
- Set the values to some value (zero, the mean, the median, etc.).
- Use imputation methods.

sklearn.impute: Impute

Survived	Pclass	Sex	Age	SibSp	Parch	Fare	Cabin	Embarked
0	3	male	22	1	0	7.25	NaN	S
1	1	female	38	1	0	71.2833	C85	C
1	3	female	26	0	0	7.925	NaN	S
1	1	female	35	1	0	53.1	C123	NaN
0	3	Male	35	0	0	8.05	NaN	S
0	3	Male	NaN	0	0	8.4583	NaN	Q
0	1	Male	54	0	0	51.8625	E46	S
0	3	Male	2	3	1	21.075	NaN	S
1	3	Female	27	0	2	11.1333	NaN	S
1	2	Female	14	1	0	30.0708	NaN	C

Contents

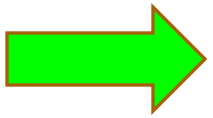
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Encoding categorical features

- Ordinal
  - May use label encoding (integer encoding)

sklearn.preprocessing.OrdinalEncoder

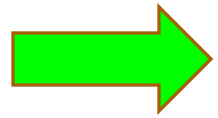
Pclass
Third class
First class
Third class
First class
Third class
Third class
First class
Third class
Third class
Second class



Pclass
2
0
2
0
2
2
0
2
2
1

category_encoders.ordinal.OrdinalEncoder

Degree
BSc
No degree
BSc
PhD
PhD
BSc
MSc
No degree
MSc
BSc



Degree
1
0
1
3
3
1
2
0
2
1

Encoding categorical features: nominal

- One hot encoding
  - Create a new binary feature to represent each of the categories

sklearn.preprocessing.OneHotEncoder

category_encoders.one_hot.OneHotEncoder

Sex	Embarked
male	S
female	C
female	S
female	S
Male	S
Male	Q
Male	S
Male	S
Female	S
Female	C

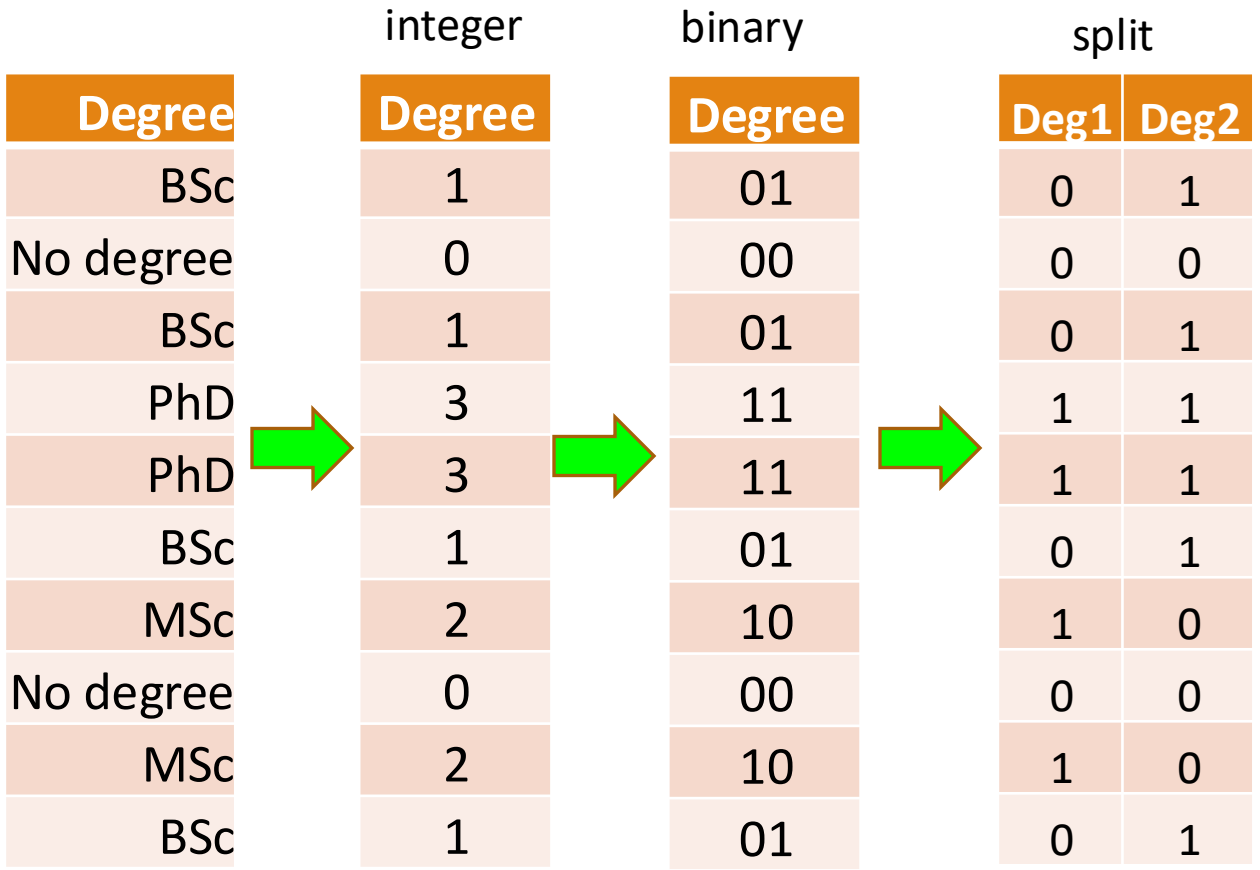


Sex		Embarked		
Female	Male	C	Q	S
0	1	0	0	1
1	0	1	0	0
1	0	0	0	1
1	0	0	0	1
0	1	0	0	1
0	1	0	1	0
0	1	0	0	1
0	1	0	0	1
1	0	0	0	1
1	0	1	0	0

Encoding categorical features: nominal

- Binary Encoding
  - The categorical feature is first converted into numerical using an ordinal encoder. Then the numbers are transformed in the binary number. After that binary value is split into different columns.
  - Binary encoding works really well when there are a high number of categories. For example, occupations.

category_encoders.binary.BinaryEncoder



E.g., a feature with upto 64 categories needs only 6 bits to encode using binary encoding.

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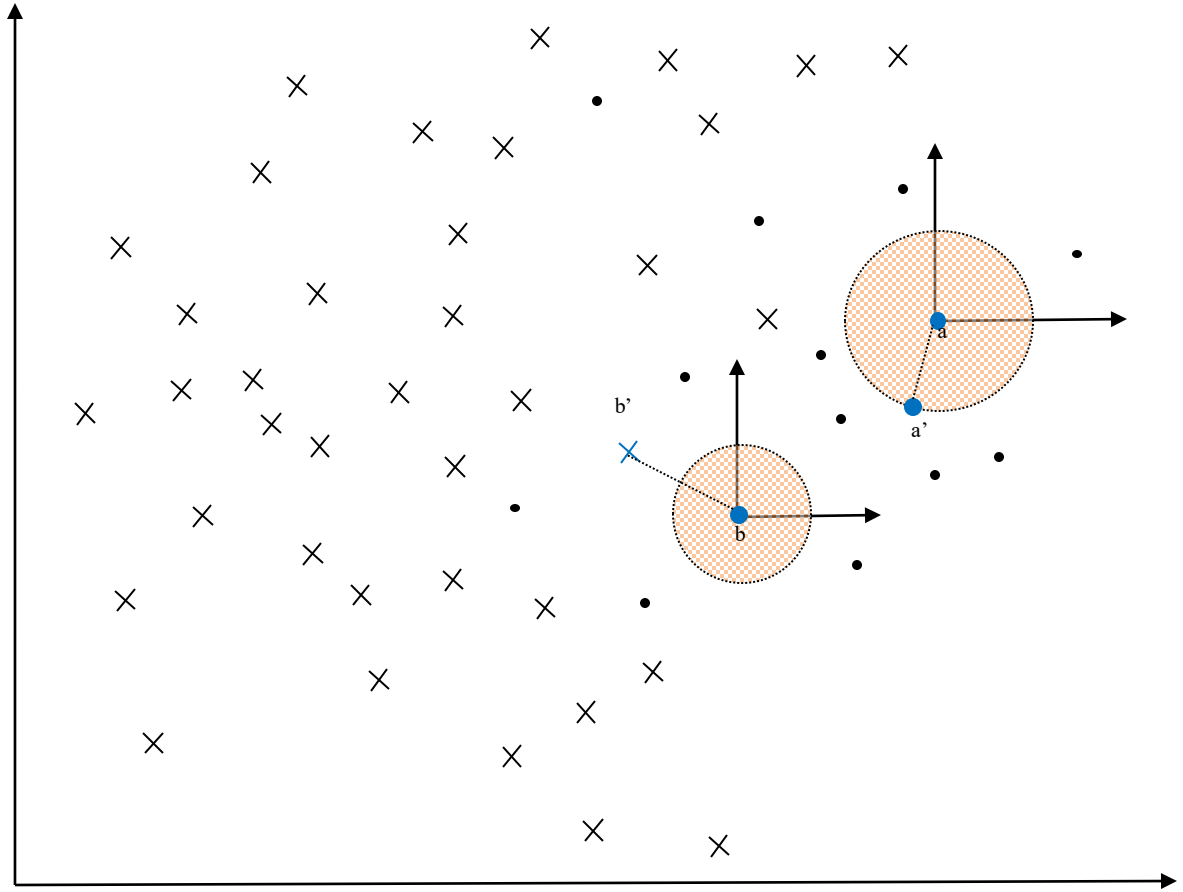
Data Preprocessing - Imbalanced data

- Imbalanced data refers to classification problems where the class distribution is not uniform among the classes.
  - Typically, they are composed by two classes: The majority (negative) class and the minority (positive) class.
- Oversampling
  - appends data to the minority class of original data set

    e.g., SMOTE (Synthetic Minority Over-sampling Technique)
- Undersampling
  - removes data from the majority class of the original data set

Imbalanced learning package: imblearn (pip install imbalanced-learn)

Imbalanced Data



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Data Preprocessing - Data reduction

- Data reduction
  - Obtains a reduced representation of the data set that is much smaller in volume but yet produce the same (or almost the same) analytical results
- Date Reduction Criteria
  - Efficiency

    The application of learning algorithm to a dataset smaller than the original one usually means a shorter computational time.
  - Accuracy

    The accuracy of the models generated represents a critical success factor, it is therefore the main criterion followed in order to select one class of learning methods over another.
  - Simplicity

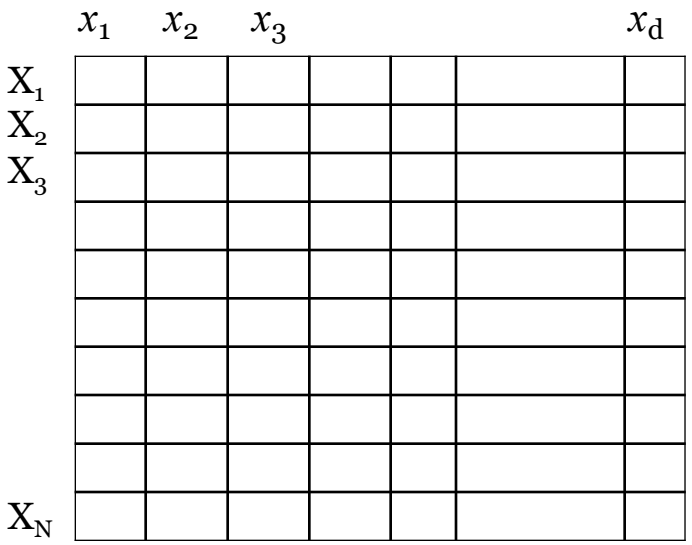
    It is important that the models generated by easily translated into simple rules that can be understood by experts in the application domain.

Data Preprocessing - Data reduction

Ways of Data Reduction

- **Feature selection** eliminates from the dataset a subset of variables which are not deemed relevant for the purpose of data modelling activities.
- **Instance selection** (Pattern selection) selects a subset of patterns from the original data that can sufficiently represent the original data in terms of some specified criteria (e.g., generalization accuracy).
- **Data transformation** transform original data into a new space so that the original data can be informatively represented in a lower dimensional space.

Ways of Data Reduction





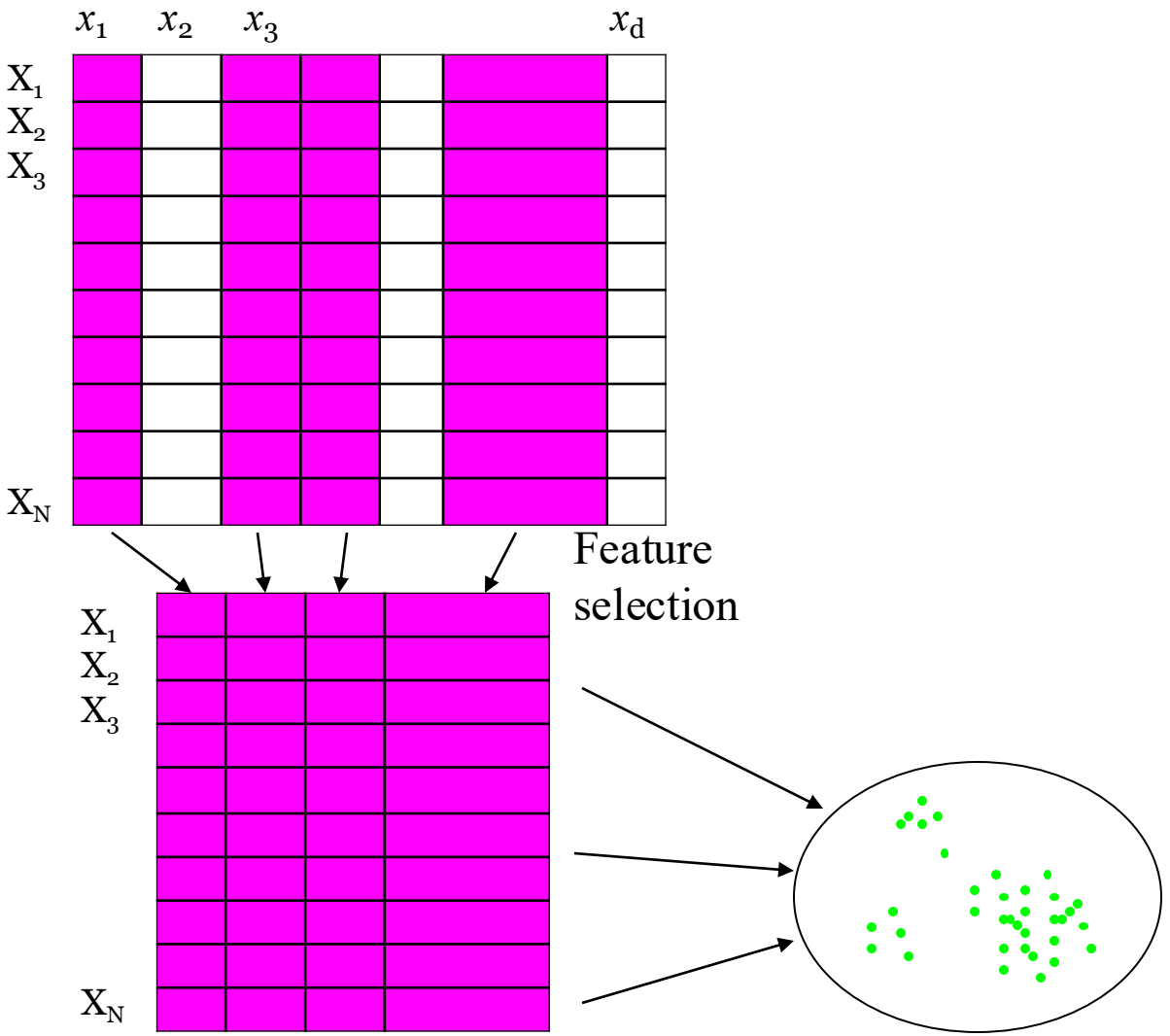
$$y = f(\mathrm{X})$$

$$\mathrm{D} = (\mathrm{X}_1 \quad \mathrm{X}_2 \quad \mathrm{X}_3 \quad \ldots\ldots \quad \mathrm{X}_N)^t$$

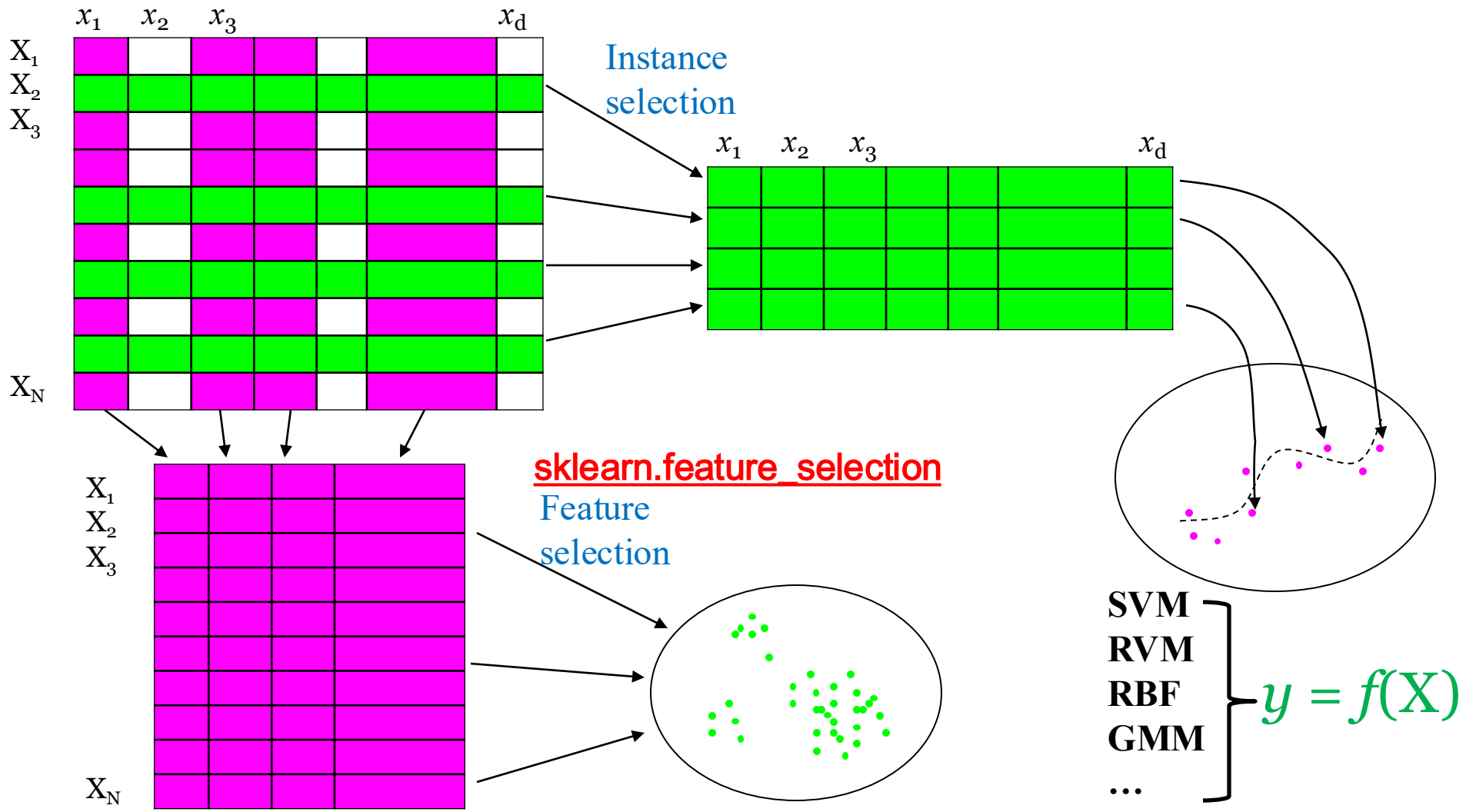
$$\mathrm{X}_i = (x_1 \quad x_2 \quad x_3 \quad \ldots\ldots \quad x_d)$$

Everything that can be counted does not necessarily count; everything that counts cannot necessarily be counted.
Albert Einstein

Ways of Data Reduction



Ways of Data Reduction



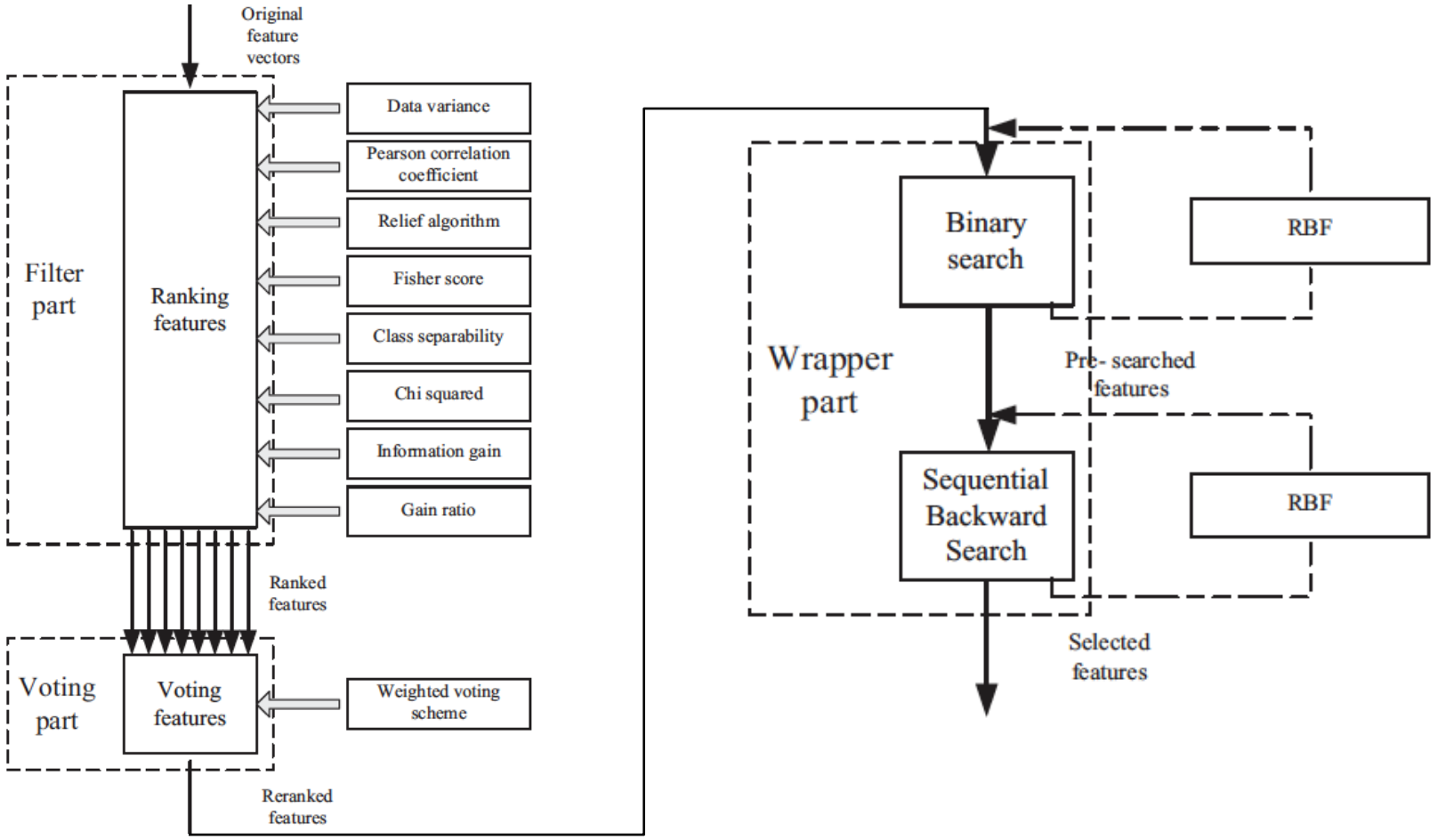
Feature selection: Methods

- Can be divided into:
  - **Unsupervised**: Make use of unlabelled data only (e.g. remove sparse or low-variance features, based on their entropy, etc.).
  - **Supervised**: Make use of the output labels, and generally are aimed at removing features that are not relevant or do not help to improve the performance of the machine learning model.

Supervised Feature Selection Methods

- Supervised feature selection methods can be further split into:
  - Filter methods: Statistical tests to score each feature.
    - Examples: Chi-squared test, correlation.
  - Embedded: Learn the most relevant features while the model is being created. Regularization is the most common technique.
    - Examples: SMLR, LASSO, Ridge Regression
  - Wrapper: Consider the selection of features as a search problem.
    - Examples: Forward/Backward selection, Recursive Feature Elimination

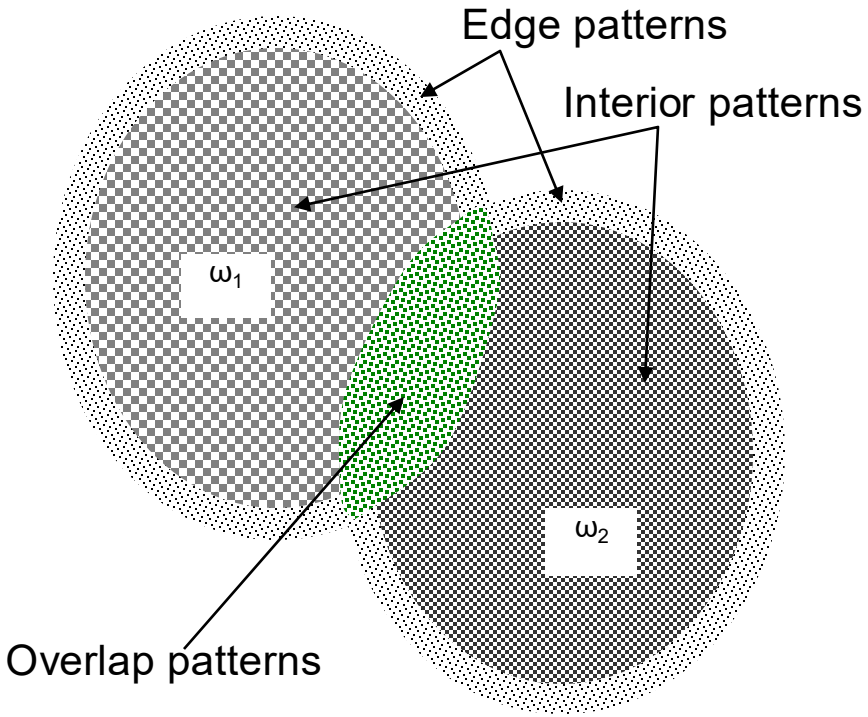
Feature Selection

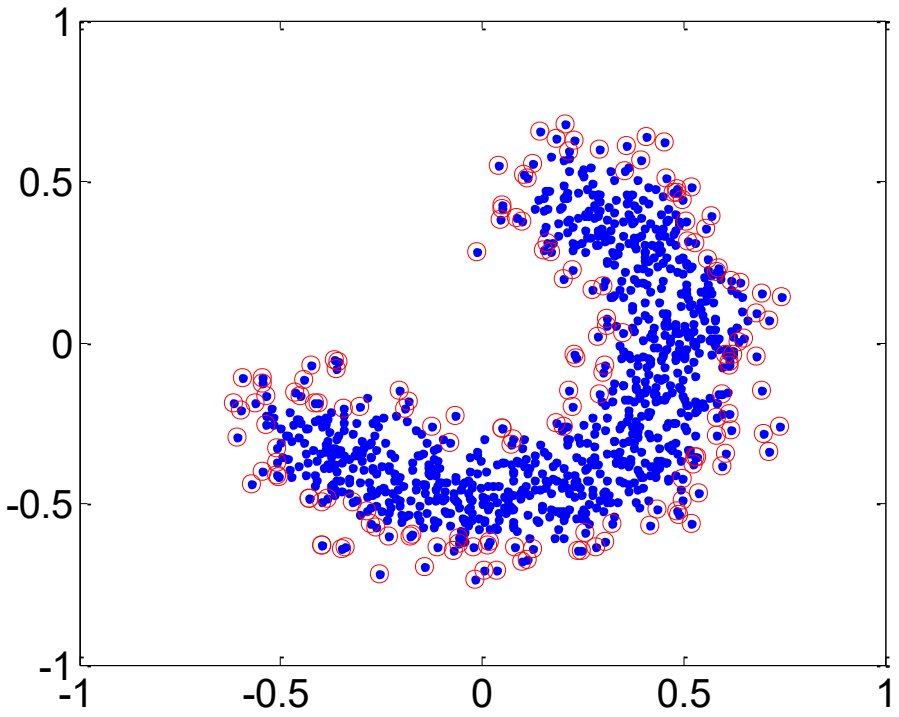


sklearn.feature_selection

Kui Zhang, Yuhua Li, Philip Scarf, Andrew Ball (2011) "Feature selection for high-dimensional machinery fault diagnosis data using multiple models and radial basis function networks," *Neurocomputing*, vol. 74, no. 17, pp. 2941-2952.

Instance/Pattern selection





Yuhua Li, Liam Maguire (2011)
"Selecting critical patterns based on local geometrical and statistical information,"
IEEE Transactions on Pattern Analysis and Machine Intelligence, vol. 33, no. 6, pp. 1189-1201.

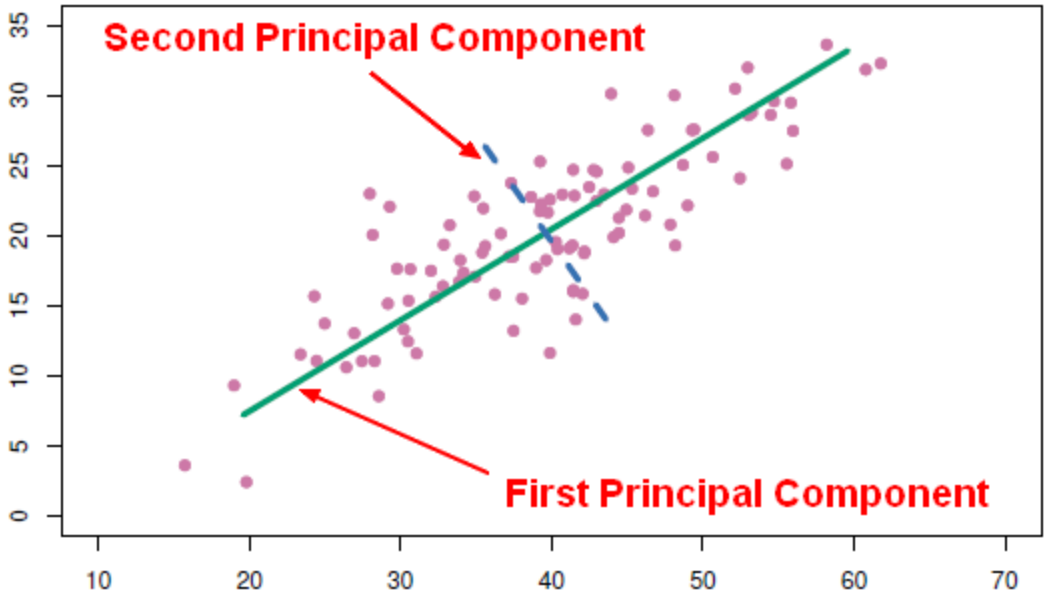
Data transformation

Dimensionality reduction

- Principal component analysis

  sklearn.decomposition.PCA

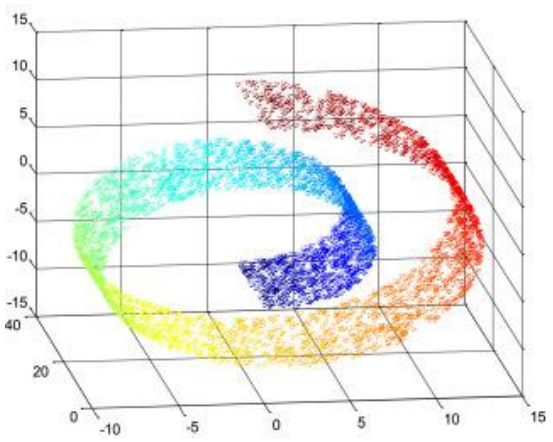
  sklearn.decomposition.KernelPCA



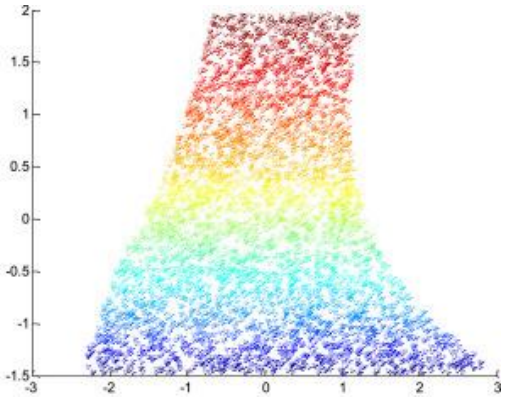
- Manifold learning
  - is an approach to non-linear dimensionality reduction

    sklearn.manifold

    UMAP: umap-learn.readthedocs.io



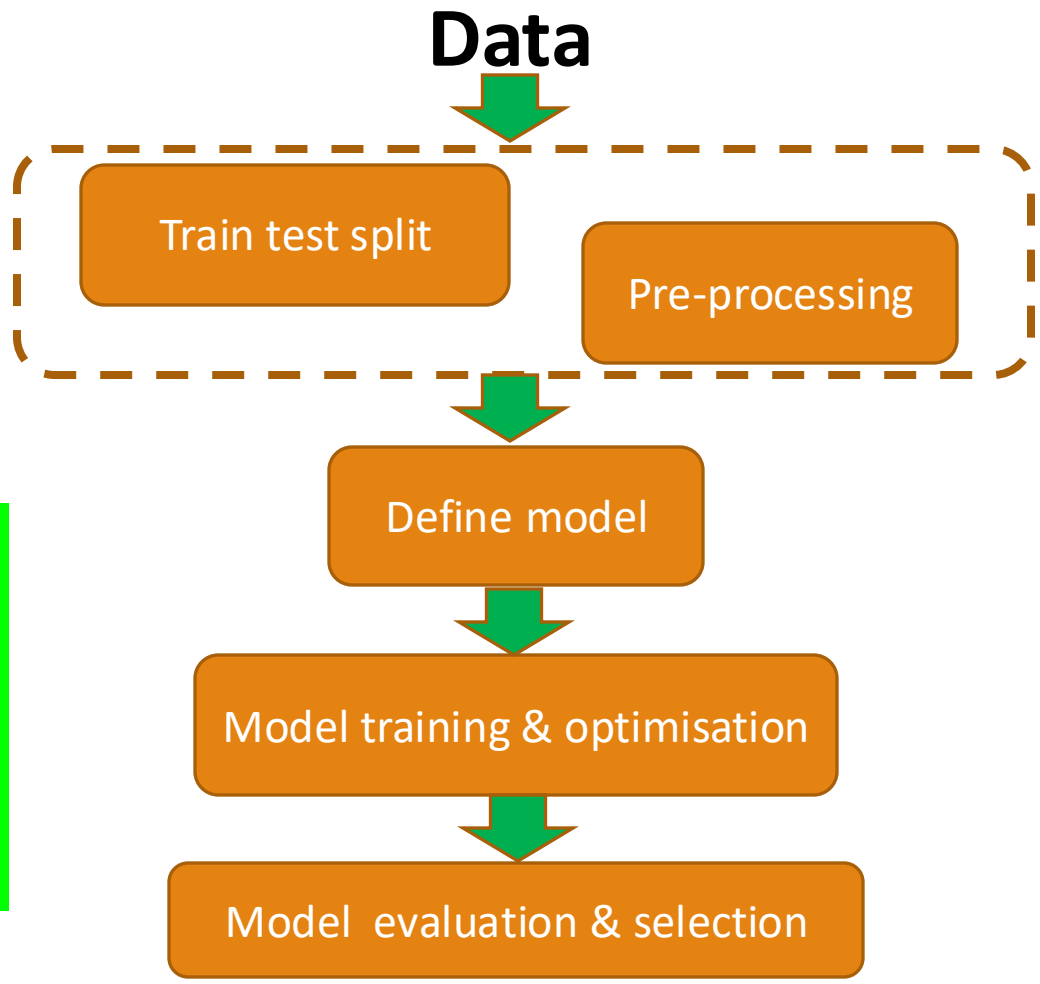
a) three-dimensional data manifold



b) two-dimensional representation of (a)

Vivien Marx , (2024) “Seeing data as t-SNE and UMAP do,” Nature Methods volume 21, pages 930–933.

Machine Learning Model in sklearn



estimator

estimator.fit()

extimator.transform()

estimator

estimator.fit()

extimator.predict()

Readings

- Chapter 2

  Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow, 3rd Edition by Aurélien Géron

- Optional reading

  Section 12.1 on PCA

  Pattern Recognition and Machine Learning, 2nd edition by Christopher Bishop

- Office hour

  4-5pm Wednesday, Abacws 4.59